

Kunal Maheshwari

Mount Pleasant, MI, USA | (989) 906-2108 | kunal.maheshwari.work@gmail.com | <https://www.linkedin.com/in/kunalmaheshwari26>

SUMMARY

OPT-EAD holder with full-time work authorization, bringing over 4 years of professional experience as a Software Developer and Data Engineer. Proven expertise in building scalable microservices, designing robust RESTful services, and developing distributed data pipelines across diverse domains. Adept in Java, Python, Spark, SQL, AWS, and Azure, with a strong command of CI/CD automation and experience in large-scale data platforms and privacy-preserving AI applications. Seeking to leverage backend development and system integration skills as a Sr Java Back End Developer.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Engineered high-performance REST and SOAP microservices using Java 8/11, Spring Boot, and Azure Functions, consistently handling over 50,000 daily API requests with 99.99 percent SLA uptime.
- Automated CI/CD pipelines leveraging Jenkins, Maven, and Azure DevOps, facilitating continuous delivery across Azure Cloud and Oracle WebLogic environments.
- Modernized legacy systems by migrating to Log4j 2.x and enhancing monitoring, decreasing issue resolution time by 20 percent.
- Spearheaded a significant EDX platform upgrade, optimizing application architecture and dependency management to achieve 50 percent faster delivery cycles.
- Collaborated within Agile and Scrum teams to deliver FHIR-compliant healthcare solutions, employing Git and SVN for version control alongside comprehensive JUnit testing.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Developed enterprise applications and RESTful web services for automotive clients using Java and EJB, focusing on backend development.
- Optimized system performance by 90 percent through strategic SQL query tuning, database enhancements, and modernization of legacy Java code.
- Implemented robust CI/CD pipelines utilizing Jenkins and GitLab, incorporating automated testing and quality gates for reliable deployments.
- Mentored junior developers through detailed code reviews, debugging sessions, and promoting clean coding practices for improved system integration.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance optimization.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and reinforcing data integrity.
- Implemented comprehensive unit testing using JUnit4, achieving 99 percent code coverage and ensuring application reliability.
- Supported release planning and sprint execution using JIRA within Agile development teams, facilitating smooth backend operations.

PROJECTS

NYU Taxi Data Analysis Platform (Spark, Python, AWS, Terraform) January 2024 - May 2025

- Developed a distributed Spark-based data pipeline to process and analyze over 8 GB of NYC taxi trip data in near real-time, enhancing backend processing capabilities.
- Built an interactive Flask web application for data exploration with dynamic filtering and visualization.
- Containerized ingestion, processing, and analytics components using Docker for reproducible deployments.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent.
- Automated cloud infrastructure provisioning using Terraform across AWS services, demonstrating infrastructure automation.

Privacy-Preserving AI Email Client (Capstone Project) (Python, Hugging Face, LoRA) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform email summarization and classification, showcasing advanced backend AI capabilities.
- Applied parameter-efficient fine-tuning using LoRA to support low-resource training under privacy constraints.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment.

SKILLS

- Programming Languages: Java, Python, SQL.
- Frameworks and Libraries: Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, LoRA, JUnit.
- Web Services: REST API, SOAP API, System Integration.
- Cloud Platforms: AWS (EC2, RDS, S3, Lambda), Azure (Functions, DevOps, ServiceBus), Google Cloud Platform.
- Databases: PostgreSQL, Oracle Database, IBM DB2, RDS.
- Tools and Technologies: Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, Terraform.
- Concepts: ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming,

CI/CD, Microservices, Large-scale data platforms, Infrastructure automation, Privacy-preserving LLM-based applications, Backend Development, Version Control.

- Data Analysis and Visualization: Tableau, QuickSight.